

Question 1 - General

- a. What do the following commands achieve when run from the command line?
- i. `git push heroku master` [2]
 - ii. `firebase login --interactive` [2]
 - iii. `mvn clean install` [2]
 - iv. `npm init` [2]
- b. Explain what the following **git** commands do when run from the current branch **master**:
- i. `git checkout -b develop` [2]
 - ii. `git checkout develop` [2]
- c. State one key difference between the two commands in b. [2]
- d. State two examples of any three elements of a basic traditional web server. [6]
- e. Virtual Host refers to the practice of running more than one application (for example, `http://www.portal.ac.org` and `http://www.assessments.ac.org`) on a single machine. The fact that they are running on the same physical server is not apparent to the end user. There are three files that must be edited when creating virtual hosts in an Apache web server.
- i. List any two such files. [2]
 - ii. Describe any two types or classes of virtual hosts. [4]

Question 2 - Web Processes

- a. List the four steps that are involved in understanding and conceptualizing software product design. [4]
- b. List any two inputs and two outputs to and from the planning and discovery phase (*requirements gathering*) of project management process. [4]
- c. Why should users be involved in all the key processes of web application development? [6]

Question 3 - Web Layers, Standards & Communication Protocols

- a. List any three standards in web programming. [3]
- b. Explain the importance of the following:
 - i. Standards in the design of web-based content. [4]
 - ii. Behavioral layer in web application development. [4]
- c. Explain any two uses of a `package.json` file in node-based applications. [4]
- d. In JavaScript, object destructuring is a concept or an approach in newer versions of the language that is used to split and access object's properties. Given the following code snippet, write a JavaScript statement or expression that can be used to destructure the `department` object into **individual** property values. [4]

```
const department = {  
  name: "Computer Science",  
  students: 4000,  
  address: {  
    city: 'Zomba',  
    addressCode: '280',  
    coordinates: [-15.3898121, 35.3351192]  
  }  
}
```

- e. Explain any three states defined by the JavaScript Promise object. [6]
- f. Using JavaScript, write a function to demonstrate the concept in 3(e) above. [3]
- g. A communication protocol is a set of rules that define & govern the communication between multiple computers over a network. These rules are what both server and client-side applications agree to maintain while communicating, various protocols exist for different application services. [2]
 - i. List any two web communication protocols. [6]
 - ii. Explain the nature of web applications that can be developed for each of the protocols listed above.

Question 4 - Web Architecture & Frameworks

a. Define the following terms:

- i. Aspect Oriented Programming (AOP) [2]
- ii. Dependency Injection (DI) [2]
- iii. Data Binding [2]
- iv. Event Propagation [2]

b. Explain the two types of data binding as used in web programming. [6]

c. Using a well labelled diagram, explain how Virtual DOM works. [6]

d. ReactJS separates the user interface (UI) into small components and nests them to build complex user interfaces. The following code snippet illustrates how that is made possible:

```
<Student {...props} >  
  <Profile></Profile>  
  <ProgramDetails></ProgramDetails>  
  <CourseList></CourseList>  
</Student>
```

Write a function based react component called **Student** that can be used to nest any number of components like in the example above. [8]

e. A number of tools exist that software engineers use for documenting and testing application programming interfaces (APIs) e.g., Swagger, Postman, Apigee. Why is software or API service documentation important? Give two reasons. [4]

Question Three: Introduction to Web Frameworks

Explain the following key words as used and applied in NestJS

- i. Injectable (2 marks)
 - ii. InjectRepository (2 marks)
 - iii. ApiTags (2 marks)
- b) Explain any three concepts defining the NestJS framework as used in web development.
(Hint: state the use of each in developing dynamic web based API services). (6 marks)
- c) Explain any three reasons why the concept of "Virtual DOM" in React is important as opposed to "Real" DOM in vanilla HTML. (6 marks)
- d) Props drilling is one way of sharing state among components.
- i. Explain the concept of "lifting up state". (4 marks)
 - ii. As a React application grows, state management becomes difficult. With examples (code snippets), explain any alternative solution for achieving proper state management. (10 marks)
- e) Explain the concept of controlled components versus uncontrolled components in React. (2 marks)
- f) Provide an example of the usage of React uncontrolled components. (2 marks)

Question Four: Web Development Processes

- a) List any two ways or tools that can be used to document software code. (2 marks)
- b) Explain any two reasons why software documentation is important in web design and development. (4 marks)
- c) Testing is a key process in web development. With the aid of a diagram, explain any three types of software tests that software developers must develop to ensure good quality of web application. (9 marks)